Pragmatic trials

- A **cluster randomised** controlled **trial** is a type of **randomised** controlled **trial** in which groups of subjects (as opposed to individual subjects) are **randomised**. ...
- What is parallel study in Clinical trials?
- A **parallel study** is a type of **clinical study** where two groups of treatments, A and B, are given so that one group receives only A while another group receives only B. Other names for this type of **study** include "between patient" and "non-crossover".

What is cluster study?

- A **cluster** randomised controlled trial is a type of randomised controlled trial in which groups of subjects are randomised.
- What is adaptive clinical trial design?
- An **adaptive design** is defined as a **design** that allows modifications to the **trial** and/or statistical procedures of the **trial** after its initiation without undermining its validity and integrity.
- The purpose is to make **clinical trials** more flexible, efficient and fast.

What is randomised cross over trials?

- A **crossover study**, or a **crossover trial**, is a longitudinal **study** in which subjects receive a sequence of different treatments (or exposures). ... In a **randomized** clinical **trial**, the subjects are randomly assigned to different arms of the **study** which receive different treatments.
- What is pragmatic study?
- to evaluate the efficacy of an intervention in a well-defined and controlled setting, whereas the term **pragmatic** was used for trials designed to test the effectiveness of the intervention in a broad routine clinical practice.

Pragmatic Approach

- Does it work?
- For whom does it work?
- How much does it cost?
- Of secondary importance – how or why does it or does not work?

Pragmatic Trials

- In the 1960s Schwarz and Llellouch coined the phrase ‘pragmatic trial’.
- In a pragmatic trial the design mimics as closely as possible ROUTINE clinical practice, with the exception that patients are randomly allocated to treatment.

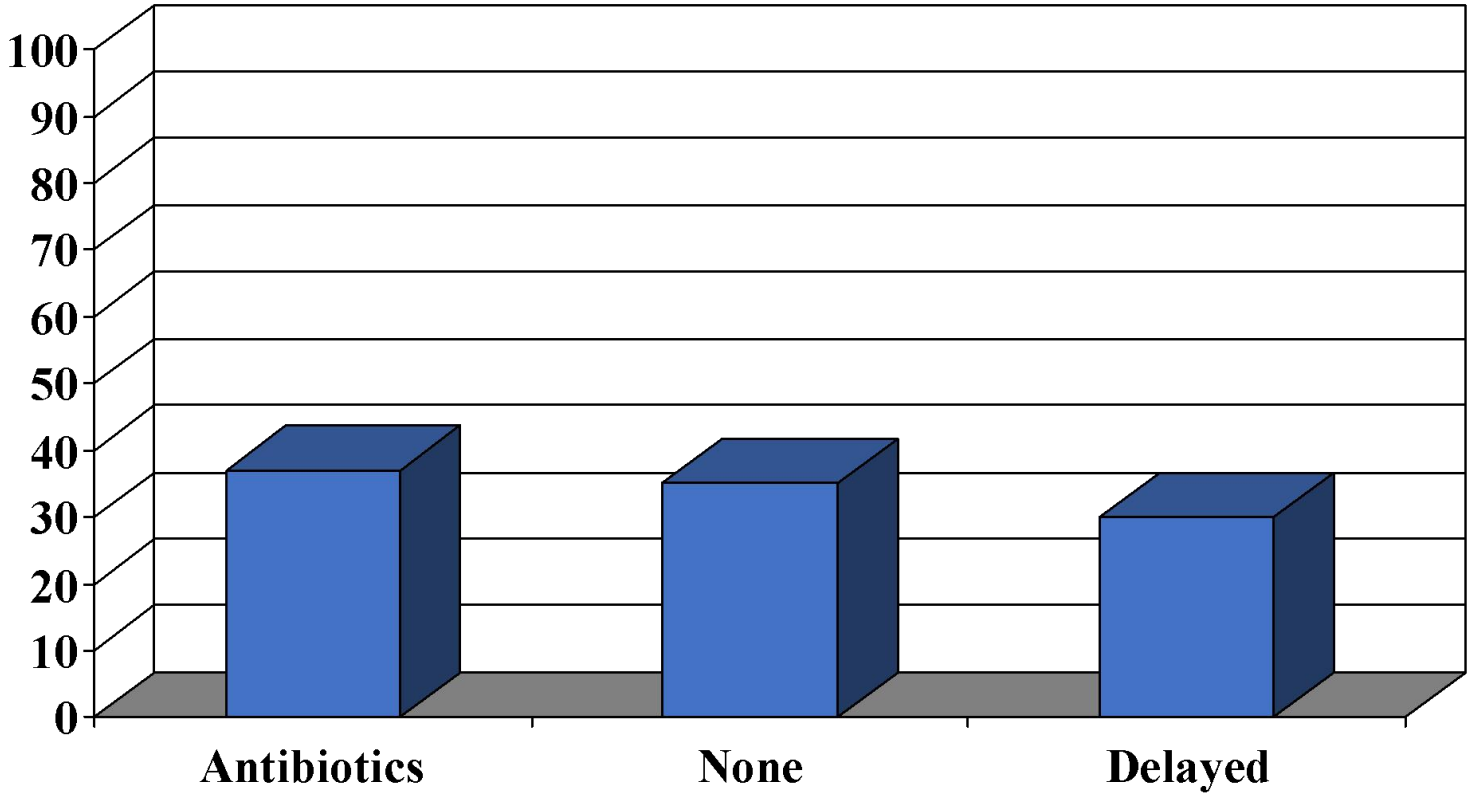
Advantages of pragmatic trials

- An advantage of the pragmatic approach is that because placebos are not used and EFFECTIVENESS is estimated.
- Because conditions mimic routine clinical practice this makes the results more applicable to the ‘average’ patient.

Antibiotics for sore throats

- The Little trial was a 3 armed trial of: immediate antibiotics; no antibiotics or delayed antibiotics. It was pragmatic BECAUSE:
  - Set in primary care where most sore throats are dealt with;
  - Did NOT use placebos;
  - Outcome was clinical severity from patient

Prevalence of Sore Throat



Antibiotics = 37%; None = 35%; Delayed = 30%

Proportion of patients better by day 3 (P=0.28)

Antibiotics

- Little *et al* showed that immediate use of antibiotics for a sore throat had NO significant effect on the resolution of symptoms

- A placebo trial would demonstrate this BUT this trial also showed that those who did not get antibiotics were less likely to visit their GP again

Pragmatic information

- GPs prescribing antibiotics for uncomplicated sore throats has very little, if any, effect

- Patients prescribed antibiotics are more likely to re-attend when they next have a sore throat compared with giving no antibiotics

Cranberry Juice for urinary tract infection

- Avorn et al randomised elderly women to receive cranberrry juice or a placebo.
- Outcomes were microbiological (I.e., bacterial counts in urine samples).
- Result – Cranberry juice significantly reduced bacteria in the urine.
- SO WHAT – what we need to know is whether it reduces symptoms.

Cranberry conclusion

- First trial was suggestive in that it showed an effect on a SURROGATE outcome measure of urinary tract infection.
- The second trial was more definite – cranberry supplementation reduces symptoms of urinary tract infection in young women.

Pragmatic with explanatory features

- Many, if not most, pragmatic trials will also include features of explanatory studies. We might measure blood pressure, cholesterol or bone mass, to elucidate mechanisms to explain why or why something did not work.
- A RCT of vitamin D injection measured a PTH levels to check the treatment was being absorbed.
- This information can be collected cheaply from pragmatic trials to enhance their scientific value.

Summary

- Explanatory trials follow the tradition of very tight control of sample – difficult to generalise from results.
- Pragmatic study much more generalisable interested in EFFECTIVENESS not too interested in how things work.
- Economic evaluations are best conducted using pragmatic studies.

What do you do?

For an patient with a painful sore throat, you wonder whether corticosteroids will help with pain relief?

- You do a search and find several studies:
  - some suggest that steroids reduce pain; some do not
- What do you do?
  - Ask a consultant? Peer? Patient?
  - Ask research student to find all studies & select the best?
- How do you know which study to believe?

You find this review

BMJ

RESEARCH

Corticosteroids for pain relief in sore throat: systematic review and meta-analysis

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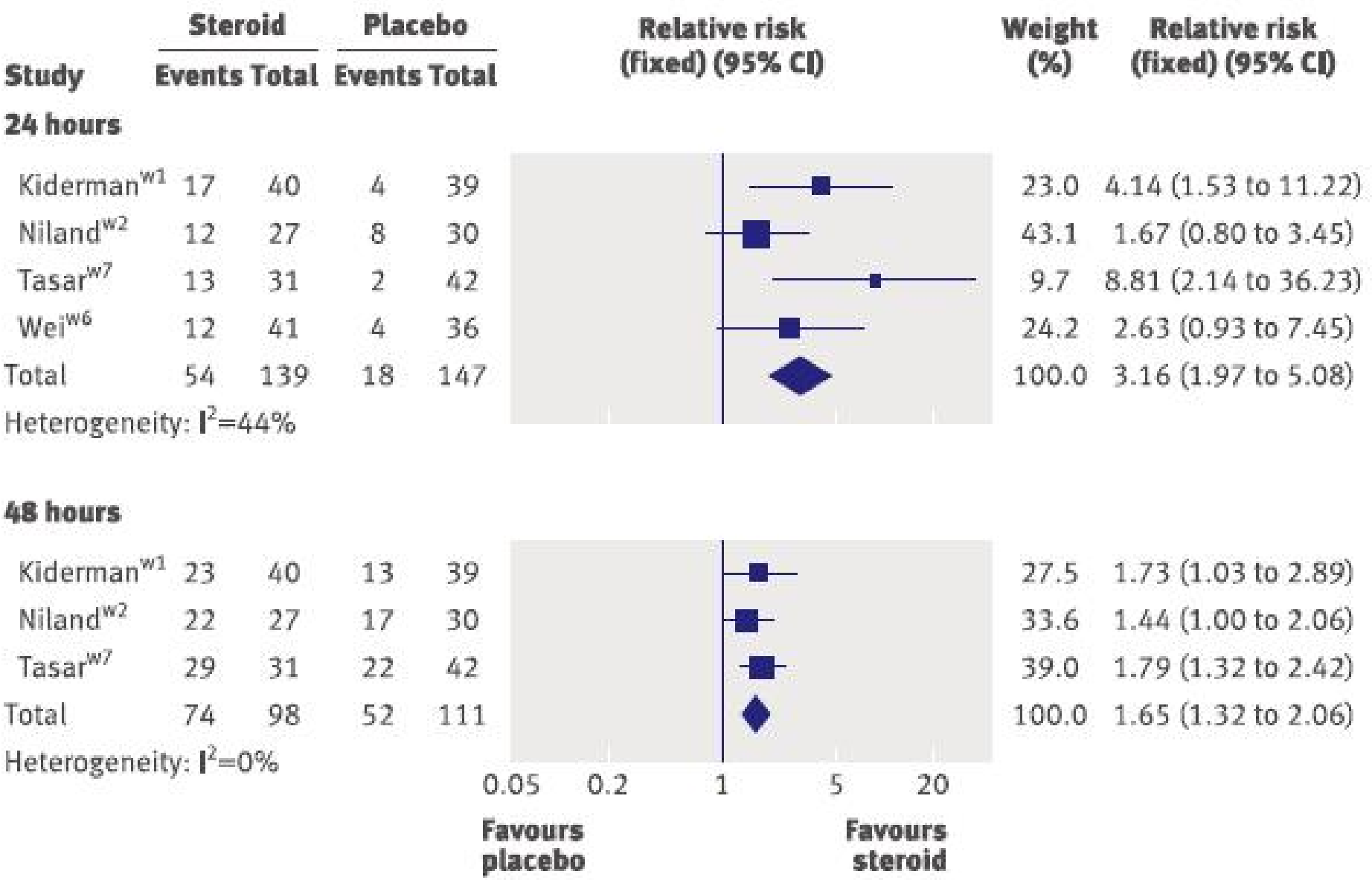


Fig 2 | Effect of corticosteroids on number of patients experiencing complete pain relief at 24 and 48 hours. See web appendix for references

How confident are you of the evidence?

WHAT IS ALREADY KNOWN ON THIS TOPIC

Corticosteroids are beneficial for symptoms of upper respiratory tract infections

Sore throat is a common condition in primary care

Recent guidelines recommend that antibiotics should not be prescribed for sore throat

WHAT THIS STUDY ADDS

At 24 hours, patients with severe sore throat who are given corticosteroids in addition to antibiotics are three times more likely to report complete resolution of symptoms than those who do not receive corticosteroids

Corticosteroids also reduce the time to mean onset of pain relief in this patient group by about 6 hours

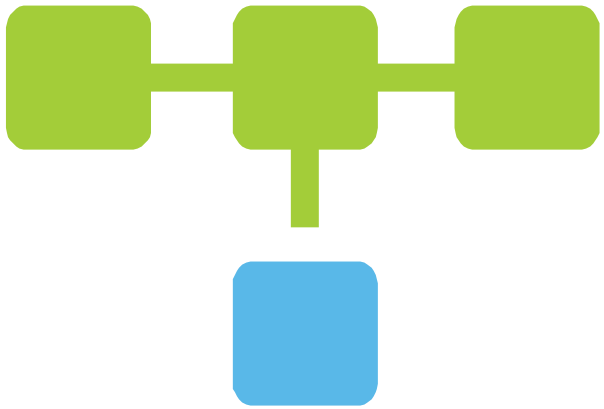
The effect of corticosteroids independent of antibiotics is unknown and should be the focus of future research

Purpose of systematic reviews

- Provide up to date summary of all published research literature
- Allow large amounts of data to be assimilated
- Provide an objective collation of results of research
- Provide reliable recommendations

Clarify the differences

- Systematic Review
- Narrative Review
- Meta-analysis



Systematic Review or meta-analysis?

- A **Systematic Review** is a review of a clearly formulated question that uses systematic and explicit methods to identify, select and critically appraise relevant research, and to collect and analyse data from the studies that are included in the review.

- Statistical methods (*meta-analysis*) may or may not be used to analyse and summarise the results of the included studies.

Narrative vs systematic review

Narrative

- Many questions
- No search methods
- No inclusion criteria
- No combining studies
- Prone to random and systematic error
- Provide conflicting summaries

Systematic

- One question
- Explicit search
  - Reproducible
- Explicit inclusion criteria
- Combine study results (meta-analysis)

WHY do we need Systematic Reviews?

Benefits of systematic reviews

- Up to date resource for clinicians
- Starting point for clinical guidelines
- Policy guidance
- Basis for new primary research
  - Important for grant funding bodies
- Management guidance

- Research training tool???

Useful Resources

- The Cochrane Collaboration www.thecochranelibrary.com/
  - Cochrane Handbook for Systematic Reviews of Interventions (version 5 updated March 2011)
- CRD www.crd.york.ac.uk/
  - The Centre for Reviews and Dissemination is a department of the University of York and is part of the National Institute for Health Research
- EPPI-Centre www.eppi.ioe.ac.uk/
  - The Evidence for Policy and Practice Information and Co-ordinating Centre, Social Science Research Unit, Institute of Education, University of London.

Steps of a systematic review

1. Clear answerable question
2. Reproducible search strategy
3. Assessment of literature quality
4. Summary of the evidence
5. Statistical, sensitivity analyses
6. Interpretation
7. Conclusions, recommendations
8. Published protocol and review

Types of systematic review

- Different research questions require different study designs →generate different types of review
- Variations occur in
  - Research questions asked
  - Primary study designs included
  - Methods for synthesis
  - Approaches to being systematic
  - Types of evidence included

Decide on process of synthesis

Factors to consider

- Consistency of outcome measures
- Sub groups
- Heterogeneity
- Common sense test

Meta analysis

- Meta-analysis and decision analysis have in common that they synthesized knowledge. Each method takes parts of the medical literature or clinical experience and, based on this information, attempts to create a whole answer to a defined problem.
- **Meta-analysis** is a qualitative and quantitative approach for systematically assessing and combining the results of previous studies in order to draw conclusions about the body of research. Studies of a topic are first systematically identified, and inclusion and exclusion criteria are defined. In traditional meta-analysis, data from the eligible studies are abstracted or collected from the investigators of the study

Advantages of meta-analysis

- Increasing sample size and power

  To detect benefits and harms of a drug or treatment,

  Good source for evidence-based clinical decision making,

  Ability to assess subgroups effects and rare events,

  Ability of indirect comparison and simultaneous evaluation of treatment therapies available for special conditions, and

  saving time, resources and money.

  cumulative meta-analysis could be as a tool to detect safety signal earlier.

Disadvantages of meta-analysis

- Susceptibility of the original studies to bias,
- publication and dissemination bias,
- practical difficulties of combining results from different studies (i.e., variations in study designs, subject characteristics, drug exposures, and outcomes),
- paradoxical results due to combining different studies,
- potential biased results from manipulation of study selection and analytic strategies, and
- challenges of including advanced study designs (e.g., case- crossover, clustered randomized trials.

Meta-analysis

- **Meta-analysis** is the statistical procedure for combining data from multiple studies. When the treatment effect (or effect size) is consistent from one **study** to the next,**meta-analysis** can be used to identify this common effect.
- **Meta-analysis** is a quantitative, formal, epidemiological **study** design used to systematically assess the results of previous **research** to derive conclusions about that body of **research**. Typically, but not necessarily, the **study** is based on randomized, controlled clinical trials.

Difference between systemic review and Meta-analysis

- A **systematic review** answers a defined research question by collecting and summarising all empirical evidence that fits pre-specified eligibility criteria.
- A **meta-analysis** is the use of statistical methods to summarise the results of these studies.

Why do we do meta-analysis

- **Meta-analysis** is a statistical technique for combining data from multiple studies on a particular topic. ... Compared to other study designs (such as randomized controlled trials or cohort studies), the **meta-analysis** comes in at the top of the 'levels of evidence' pyramid in evidence-based healthcare.
- Advantages:
- a meta-analysis uses a statistical approach to combine the results from multiple studies in an effort to increase **power** (over individual studies), improve estimates of the size of the effect and/or to resolve uncertainty when reports disagree.

Pharmacoeconomic research

- Pharmacoeconomic research considers both costs and utilities of outcome include
- **cost-minimization,**
- **cost-effectiveness,**
- **cost-utility, and**
- **cost-benefit.**

Cost-minimization analysis

- **Cost-minimization analysis** (or **cost-identification** analysis) assumes that alternative treatment result in the same outcome, and value or decision is assigned entirely based on the lowest cost. It is appropriate only if treatment outcomes or benefits are equivalent among the therapies being evaluated.
- Ex. Evaluation of brand name Vs generic (with same outcome or indication)

Cost-benefit analysis

- **Cost-benefit analysis** remains a less common used technique in economic assessments of health care in the US, because it is difficult to translate all outcomes into monetary measures.
- EX. Pharmacist managed a clinic for a disease Vs Cardiac rehabilitation program.

Cost-effectiveness analysis

- **Cost-effectiveness analysis** assumes that cost and the effectiveness of interventions vary, and uses natural units as outcome measures. Consequently, value is assigned by the cost per unit improvement of a health status, such as dollars per years of life saved.
- Cost/ mm Hg decrease of Blood pressure in the treatment

Cost-utility analysis

- **Cost-utility analysis,** a type of cost-effectiveness analysis, often presents the outcome as cost per quality-adjusted life-year. The advantage of cost-utility analysis is that the utility assignment can be generalized to all health states.
- Ex. Cost/ Quality adjusted life year gained for aspirins and statins or both drugs for prevention of coronary heart disease in man.